

CURRICULUM VITAE

MASATARO ASAI, PH.D

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RESEARCH INTEREST

Neuro-Symbolic Systems (Machine Learning + Classical AI) for search/planning/reasoning, with an emphasis on Bayesian generative modeling (e.g., variational inference). This includes: (1) **Automatic identification of discrete structured representation for planning**, i.e., symbol grounding and knowledge acquisition. (2) **Learning a search policy / heuristics** for faster problem solving. (3) **Statistically well-motivated search algorithms** inspired by multi-armed bandits. *Colors in the bibliography denote each theme.* Recent work also focuses on LLM-based reasoning, inference-time scaling, and algorithmic guarantees while using LLMs.

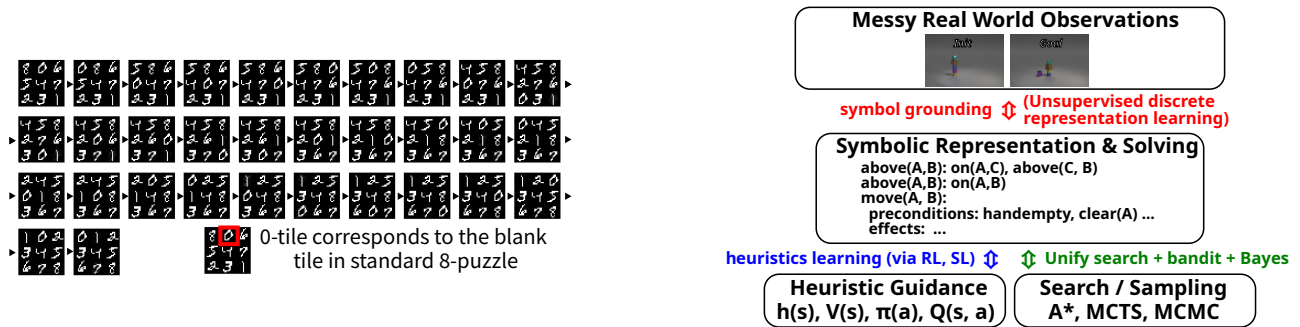


Figure 1: **(Left)** Result of visually solving 8-puzzle using a learned discrete latent representation and a classical planner. **(Right)** Cognitive system can have neural components for three distinct purposes: **Perception**, **heuristics**, and **search**.

WORK EXPERIENCE

- 07/2019–present.** Staff Research Scientist (PI) at **MIT-IBM Watson AI Lab, IBM Research Cambridge** (Full Time).
- 04/2018–07/2019.** Research Staff Member (PI), Embodied Learning Group, **IBM Research Tokyo** (Full Time).
- 08/2016–11/2016** Research Intern, **IBM Research Ireland**.
- 04/01/2016–03/31/2018** Research Fellow (DC2), Japan Society for the Promotion of Science.

AWARDS

- 10/2024** Outstanding Paper Award, 27th European Conference on Artificial Intelligence.
- 06/2022** ICAPS 2022 Outstanding PC and SPC Award (Outstanding Program Committee Members).
- 03/2017** JSAI Annual Conference Student Award, The Japanese Society for Artificial Intelligence.

GRANTS

- 09/2022-09/2024** DTIC contract FA8075-18-D-0008, Task Order FA807520F0060, Task 4 - Autonomous Defensive Cyber Operations (DCO) Research & Development (R&D).
- 04/2016-03/2018** JSPS Grant-in-Aid for JSPS Fellows and a JSPS KAKENHI grant, Project No. 16J10073.

THESIS SUPERVISION

- 2024** Carlos Núñez-Molina*, Universidad de Granada (Ph.D)
- 2025** Amin Seffo, Technical University of Munich, (M.Sc)
- 2025** Supriya Lall, MIT EECS (M.Eng)

MOST RECENT PUBLICATIONS / PREPRINTS

- [1] Masataro Asai. **Bilevel MCTS for Amortized O (1) Node Selection in Classical Planning**. In *Proc. AAAI Conference on Artificial Intelligence (CORE Rank: A*)*, 2026.
- [2] Masataro Asai and Stephen Wissow. **Extreme Value Monte Carlo Tree Search for Classical Planning**. In *Proc. AAAI Conference on Artificial Intelligence (CORE Rank: A*)*, 2026.
- [3] Aladin Djuhera, Amin Seffo, Masataro Asai, and Holger Boche. **“Don’t Do That!”: Guiding Embodied Systems through Large Language Model-based Constraint Generation**. *arXiv:2506.04500 (under submission @ ICRA 2026)*, 2025.
- [4] Supriya Lall, Christian Farrell, Hari Pathanjaly, Marko Pavic, Sarvesh Chezian, and Masataro Asai. **Verbalized Algorithms**. In *NeurIPS 2025 Workshop on Efficient Reasoning*, 2025.
- [5] Miguel Tulla, Andrea Vignali, Christian Colon, Giancarlo Sperli, Simon Pietro Romano, Masataro Asai, Una-May O’Reilly, and Erik Hemberg. **ALFA-Chains: AI-Supported Discovery of Privilege Escalation and Remote Exploit Chains**. *arXiv:2504.07287 (under submission @ Network and Distributed System Security (NDSS) Symposium 2026)*, 2025.

CONFERENCE PUBLICATIONS (*: PRESENTER) / JOURNAL PUBLICATIONS / BOOK CHAPTERS

- [1] Carlos Núñez-Molina*, Masataro Asai, Pablo Mesejo, and Juan Fernández-Olivares. **On Using Admissible Bounds for Learning Forward Search Heuristics**. In *Proc. International Joint Conference on Artificial Intelligence (IJCAI) (CORE Rank: A*)*, 2024.
- [2] Stephen Wissow and Masataro Asai*. **Scale-Adaptive Balancing of Exploration and Exploitation in Classical Planning**. In *European Conference on Artificial Intelligence (ECAI) (CORE Rank: A)*, 2024. **Outstanding Paper Award**.
- [3] Masataro Asai and Benjamin Ayton*. **Is Policy Learning Overrated?: Width-Based Planning and Active Learning for Atari**. In *Proc. International Conference of Automated Planning and Scheduling (ICAPS) (CORE Rank: A*)*, 2022.
- [4] Masataro Asai, Clement Gehring*, Rohan Chitnis, Tom Silver, Leslie Pack Kaelbling, Shirin Sohrabi, and Michael Katz. **Reinforcement Learning for Classical Planning: Viewing Heuristics as Dense Reward Generators**. In *Proc. International Conference of Automated Planning and Scheduling (ICAPS) (CORE Rank: A*)*, 2022.
- [5] Masataro Asai, Hiroshi Kajino, Alex Fukunaga, and Christian Muise. **Classical Planning in Deep Latent Space**. *J. Artif. Intell. Res. (JAIR) (CORE Rank: A)*, 2022.
- [6] Masataro Asai, Hiroshi Kajino, Alex Fukunaga, and Christian Muise. **Symbolic Reasoning in Latent Space: Classical Planning as an Example**. In Pascal Hitzler and Md. Kamruzzaman Sarker, editors, *Neuro-Symbolic Artificial Intelligence: The State of the Art*, volume 342 of *Frontiers in Artificial Intelligence and Applications*, page 52–77. IOS Press, 2021.
- [7] Masataro Asai* and Christian Muise. **Learning Neural-Symbolic Descriptive Planning Models via Cube-Space Priors: The Voyage Home (to STRIPS)**. In *Proc. International Joint Conference on Artificial Intelligence (IJCAI) (CORE Rank: A*)*, 2020.
- [8] Masataro Asai*. **Unsupervised Grounding of Plannable First-Order Logic Representation from Images**. In *Proc. International Conference of Automated Planning and Scheduling (ICAPS) (CORE Rank: A*)*, 2019.
- [9] Masataro Asai* and Hiroshi Kajino. **Towards Stable Symbol Grounding with Zero-Suppressed State AutoEncoder**. In *Proc. International Conference of Automated Planning and Scheduling (ICAPS) (CORE Rank: A*)*, 2019.
- [10] Masataro Asai* and Alex Fukunaga. **Classical Planning in Deep Latent Space: Bridging the Subsymbolic-Symbolic Boundary**. In *Proc. AAAI Conference on Artificial Intelligence (CORE Rank: A*)*, 2018.
- [11] Masataro Asai* and Alex Fukunaga. **Exploration Among and Within Plateaus in Greedy Best-First Search**. In *Proc. International Conference of Automated Planning and Scheduling (ICAPS) (CORE Rank: A*)*, 2017.
- [12] Masataro Asai and Alex Fukunaga. **Tie-Breaking Strategies for Cost-Optimal Best First Search**. In *J. Artif. Intell. Res. (JAIR) (CORE Rank: A)*, volume 58, page 67–121, 2017.

- [13] Masataro Asai*, Akihiro Kishimoto, Adi Botea, Radu Marinescu, Elizabeth M. Daly, and Spyros Kotoulas. Efficient Optimal Search under Expensive Edge Cost Computation. In *Proc. International Joint Conference on Artificial Intelligence (IJCAI) (CORE Rank: A*)*, 2017.
- [14] Masataro Asai* and Alex Fukunaga. [Tiebreaking Strategies for A* Search: How to Explore the Final Frontier?](#) In *Proc. AAAI Conference on Artificial Intelligence (CORE Rank: A*)*, 2016.
- [15] Masataro Asai* and Alex Fukunaga. [Solving Large-Scale Planning Problems by Decomposition and Macro Generation](#). In *Proc. International Conference of Automated Planning and Scheduling (ICAPS) (CORE Rank: A*)*, 2015.
- [16] Masataro Asai* and Alex Fukunaga. [Fully Automated Cyclic Planning for Large-Scale Manufacturing Domains](#). In *Proc. International Conference of Automated Planning and Scheduling (ICAPS) (CORE Rank: A*)*, 2014.

WORKSHOP AND OTHER PUBLICATIONS (*: PRESENTER)

- [1] Masataro Asai and Stephen* Wissow. [Extreme Value Monte-Carlo Tree Search \(Extended Abstract\)](#). In *Proc. of Annual Symposium on Combinatorial Search*, 2024.
- [2] Masataro Asai. [Analytical Conjugate Priors for Subclasses of Generalized Pareto Distributions](#). *arXiv:2303.12199*, 2023.
- [3] Masataro Asai. Dr. Neurosymbolic, or: How I Learned to Stop Worrying and Accept Statistics. *CoRR*, abs/2209.04049, 2022.
- [4] Masataro Asai. [Generating Plannable Lifted Action Models for Visually Generated Logical Predicates](#), 2021.
- [5] Masataro Asai*. [Neural-Symbolic Descriptive Action Model from Images: The Search for STRIPS](#). In *Proc. ICAPS Workshop on Knowledge Engineering for Planning and Scheduling (KEPS)*, 2020.
- [6] Masataro Asai and Zilu Tang. [Discrete Word Embedding for Logical Natural Language Understanding](#). In *Proc. ICAPS Workshop on Knowledge Engineering for Planning and Scheduling (KEPS)*, 2020.
- [7] Masataro Asai. Set Cross Entropy: Likelihood-based Permutation Invariant Loss Function for Probability Distributions. *CoRR*, abs/1812.01217, 2019.
- [8] Masataro Asai*. [Diversification Mechanisms for Best-First Search](#). PhD thesis, Department of General Systems Studies, Graduate School of Arts and Sciences, The University of Tokyo, 2018.
- [9] Masataro Asai. [Photo-Realistic Blocksworld Dataset](#). *CoRR*, abs/1812.01818, 2018.
- [10] Shoma Endo, Masataro Asai*, and Alex Fukunaga. Evaluation of a Simple, Window-based, Replanning Approach to Plan Optimization. In *Proc. ICAPS Workshop on Heuristics and Search for Domain-independent Planning (HSDIP)*, 2016.
- [11] Masataro Asai* and Alex Fukunaga. Applying Problem Decomposition to Extremely Large Planning Domains. In *Proc. ICAPS Workshop on Knowledge Engineering for Planning and Scheduling (KEPS)*, 2014.

PATENTS

Discrete feature representation with class priority, US Patent App. 16/290250
 Permutation-invariant optimization metrics for neural networks, US Patent App. 16/366678
 Discovering higher-level actions from expert’s action demonstration, US Patent App. 16/419183
 Symbolic model training with active learning, US Patent App. 17/132776

EDUCATION

04/2015–03/2018 *Ph.D.* Artificial Intelligence, Heuristic Search, Planning, Scheduling, Optimization. Thesis title: *Diversification Mechanisms for Best-First Search*; Advisor: A. Fukunaga
04/2013–03/2015 *M.A.* Artificial Intelligence, Heuristic Search, Planning, Scheduling, Optimization. Thesis title: *Automated Cyclic Planning for Large Scale Planning Problems*; Advisor: A. Fukunaga

04/2009–03/2013 B.Eng in Traffic Simulation. Multi-Agent Model, Spatial Search. Thesis title: *Distributed Cooperative Agents in Microscopic Traffic Simulation using St-RRT*; Advisor: S. Yoshimura. H. Fujii.

LANGUAGE

English: TOEFL 105/120 (Reading:29/30, Listening:29/30, Speaking:22/30, Writing:25/30, Dec 2014).

Japanese: Native

TECHNICAL SKILL

Programming Paradigm: Object-Oriented / Functional / Logic / Rule-based programming, Metaprogramming, low-level assembly optimization, Domain Specific Language (DSL) development, compile-time optimization.

Development: Git, GitHub Flow, Test-Driven Development and Continuous Integration (Travis-CI / CircleCI).

Languages: (Professional) Common Lisp, C++, Bash, Python, Javascript, (Intermediate) C, Java, (Elementary) Ruby

Frameworks: Pytorch, TensorFlow/Keras, Containerization (Docker), Cloud (Amazon AWS), Cluster management (Torque/PBS, LSF), Node.js

INVITED TALKS / COMMUNITY SERVICES / OTHER ACTIVITIES

(2025) OSS contributions to huggingface and vllm.

(2024/09/05-06) Invited talk at SIG-FPAI (Special Interest Group on Fundamental Problems in Artificial Intelligence), Iwakuni, Japan. https://sig-fpai.org/past/fpai129_cfp.html

(2021/07/15) Invited talk at RSS 2021 Workshop on Declarative and Neurosymbolic Representations in Robot Learning and Control, in 2021 Robotics: Science and Systems (RSS) Conference. <https://dnr-rob.github.io/>

(2019/12/10) Featured in Nikkei Business Publications <https://xtech.nikkei.com/atcl/nxt/mag/rob/18/012600001/00044/>

(2016-) PC / SPC member roles in AI conferences (AAAI, IJCAI, ICAPS, AAMAS, ICRA, Neurips, ICLR, ICML)

(2019) numcl, A Numpy clone in Common Lisp. Was temporarily #5 on Hacker News. 600+ stars.

(2015) eazy-opencl : Common Lisp interface to OpenCL 2.0 (GPGPU language similar to CUDA).

(2015) trivia : An extensible and fast pattern matching compiler in Common Lisp. 300+ stars.

(2013–2018) Compute cluster maintainance and management (80 cores) with NFS/NIS/Torque-PBS. Live monitoring/power consumption management. Secure VPN network over the campus.

(2011–2012) Mechanical engineering under project professor Kohei Kusaka (former World Rally Championship co-driver): Full engine modification & rebuilding of 1.8 liter Mazda BP engine, fuel map / ignition timing optimization, map visualization and variable resonance intake system (Arduino).